Migrating VMware Workloads to Azure Local with Azure Migrate

Discovery, replication, migration, verification and completion for moving VMware virtual machines to an Azure Local instance with Azure Migrate. The assessment that feeds it (inventory, sizing, waves) is covered in the sizing guide.

Status: prepared 2026-10-06 and re-checked 2026-10-13 against Microsoft Learn (2609 release family; the run table follows the current replicate and migrate articles, which have no test-migration step for Azure Local targets). Migration of **VMware** VMs to Azure Local is generally available (announced October 2025); migration of **Hyper-V** VMs is a **preview**. Both statuses were checked against the 2609 documentation on 6 Oct 2026. Status and limits change, so re-read the migration overview for your release when you plan.

What you need in place

Component	Requirement
Target	An Azure Local instance (2311.2 or later; the migration articles apply to 2503 and later), deployed, registered and validated; its custom location noted; a storage path and a logical network for the migrated VMs
Source	vCenter Server and ESXi 8.0, 7, 6.7 or 6.5
Source appliance	A Windows Server 2022 or 2025 VM on the VMware side with at least 16 GB memory, 80 GB disk and 8 vCPUs
Target appliance	A Windows Server 2022 or 2025 VM on Azure Local, same size
Network	The source environment must be able to open a network connection to the target Azure Local instance, on the same network or over a VPN
Appliances and registration	Each appliance registers to the Azure Migrate project; use a separate Microsoft Entra application per appliance if you use preconfigured applications

Firewall ports (Microsoft Learn): on the source side 443 (Azure Migrate services and vCenter), 902 (replication from ESXi snapshots), 445 (SMB between source and target appliance), 3389 and 44368 (appliance access); on the target side 445, 5985 and 5986 (WinRM from the appliance to the host), 3389 and 44368. Allow the Azure Migrate URLs and `*.siterecovery.azure.com`.

VDDK: Azure Migrate does not provide the VMware VDDK packages and access to them can be restricted. If a supported VDDK package is not available to your organization, use a partner migration option instead.

Prepare the source VMs

- Bring all disks online and persist drive letters (configure the SAN policy), or the disks come up offline after migration.
- Disable BitLocker on Windows VMs; encrypted disks and volumes are not supported.
- Remove shared disks; they are not supported.
- Linux VMs: make sure the Hyper-V Linux Integration Services are in the init image so the VM boots on Azure Local.

- Check Secure Boot on UEFI VMs; the setting is preserved. If you change it, allow up to 30 minutes for discovery to see the change.
- If a VM already has the Azure Connected Machine agent, decide whether to reuse its Arc resource (keep the agent) or uninstall the agent before replication.
- Snapshot-based backup of the source VMs must not run during replication cycles.
- The boot type is preserved: BIOS VMs become Hyper-V generation 1 VMs and UEFI VMs become generation 2. Note the generation 1 limitations in the Azure Local VM management documentation.

Plan the waves

1. Start from the assessment: complexity score, dependencies, recovery objectives and owners per workload.
2. Group VMs that depend on each other in one wave; one wave is one change window.
3. Select up to 10 machines at a time for replication; add further machines in batches of 10. One appliance replicates 52 disks in parallel, so large waves queue.
4. Record the missing evidence. Aggregate fit and an inventory export do not establish application dependencies; obtain them from dependency analysis or the application owner and record anything else as an assumption.

Run a migration

Step	Action	Evidence
1	Create the Azure Migrate project in the Azure portal; deploy the source appliance (project key, VDDK, vCenter account; leave agentless dependency analysis off, it is not supported for Azure Local); onboard to Azure Local; discover	VMs listed with sizes and readiness
2	Deploy the target appliance on the Azure Local instance	Appliance shows connected
3	Replicate (up to 10 VMs per batch): choose the target logical network and storage path, vCPU and memory, VHDX disks	Initial copy into seed disks, then hourly deltas; status Ready to migrate
4	Migrate: shut down the source (recommended), planned failover with a final sync, start the target	VM shows as an Azure Local VM enabled by Azure Arc; status Completed
5	Verify in the portal, the guest and the application (there is no test-migration step for Azure Local targets, so this is the check)	VM running, source off, disks and network as configured; application owner sign-off
6	Complete migration	Delete protected item job done: replication state and seed disks removed, the migrated VM untouched
7	Enable guest management on the migrated VM	Extensions, monitoring, updates and backup can be applied

To migrate the same source VM again later, complete the migration on the migrated resource first; that clears the state and allows replication to be enabled again.

Rollback: the source VM is the rollback. Keep it powered off, not deleted, until the verification window closes.

After the cutover

A migrated VM is an Azure Local VM managed through Azure Arc. Bring it inside the Day-2 scope: monitoring, update management, backup, access control and the governance baseline. Tag it so dynamic scopes and policy pick it up. Confirm it is protected before the source VM is retired.

Common problems

Symptom	Cause	Action
Disks offline after migration	SAN policy not set on the source	Bring disks online in the guest; set the SAN policy on the remaining sources
Linux VM does not boot	Hyper-V drivers missing from the init image	Rebuild the init image with Hyper-V drivers on the source
Replication stalls	Source snapshot backup running, or port 902 blocked	Reschedule backup; check ports
Cannot select the VM for replication	BitLocker enabled, shared disk, or encrypted volume	Remove the blocker on the source
Migrated VM has no network	Logical network or IP pool not prepared	Create the logical network and pool before replication

Migrating from Hyper-V (preview)

The same pattern applies with a Hyper-V source: Windows Server 2012 R2 to 2022 hosts are discovered, and the Hyper-V generation is preserved. This route is a **preview**; do not plan production waves on it without confirming its status.